Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (Practical) (CS604P)

Question: 7 (Marks: 1)

_____ is also used for creating the Fifo in Unix/Linux environment.

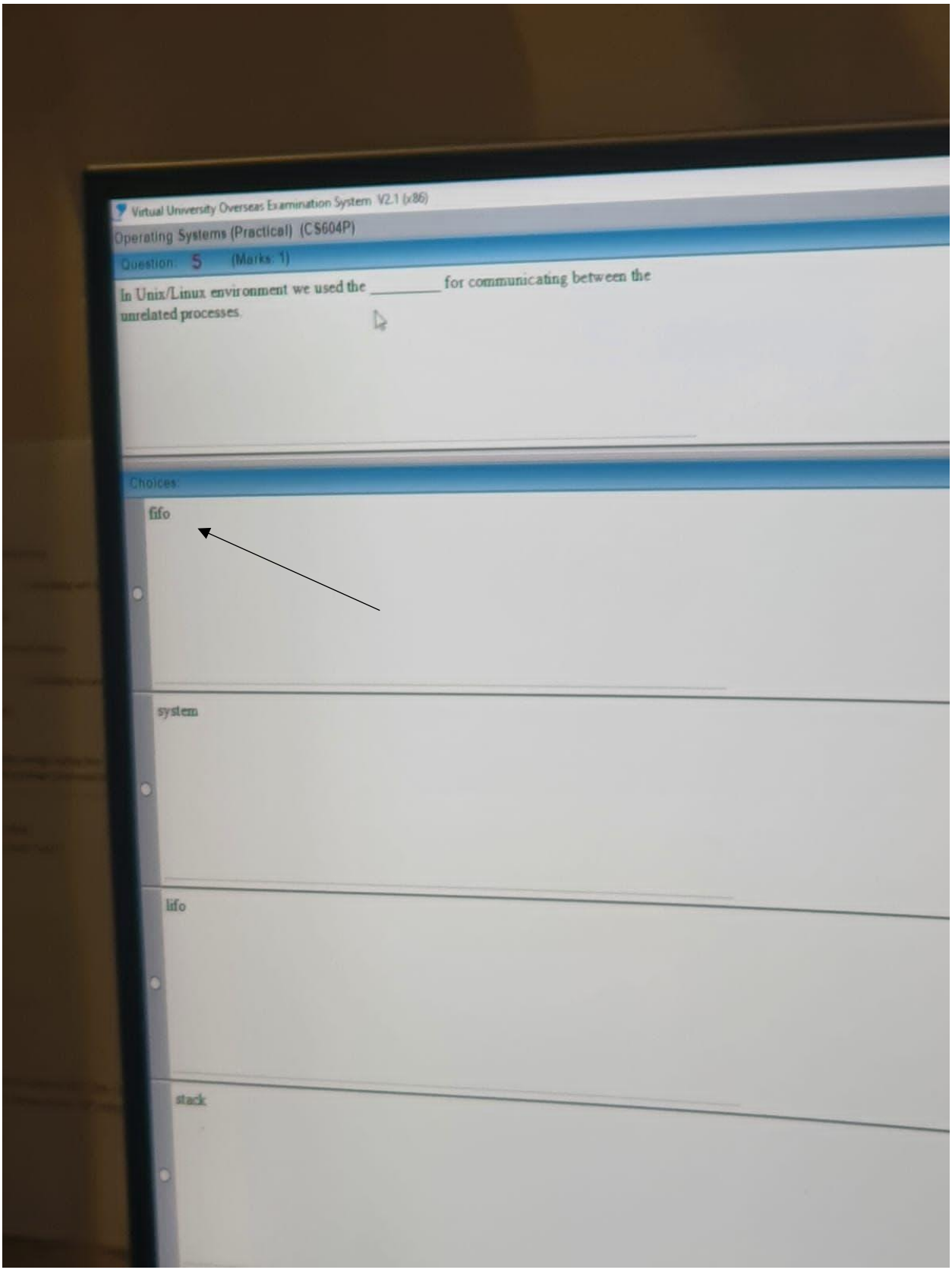
Choices:

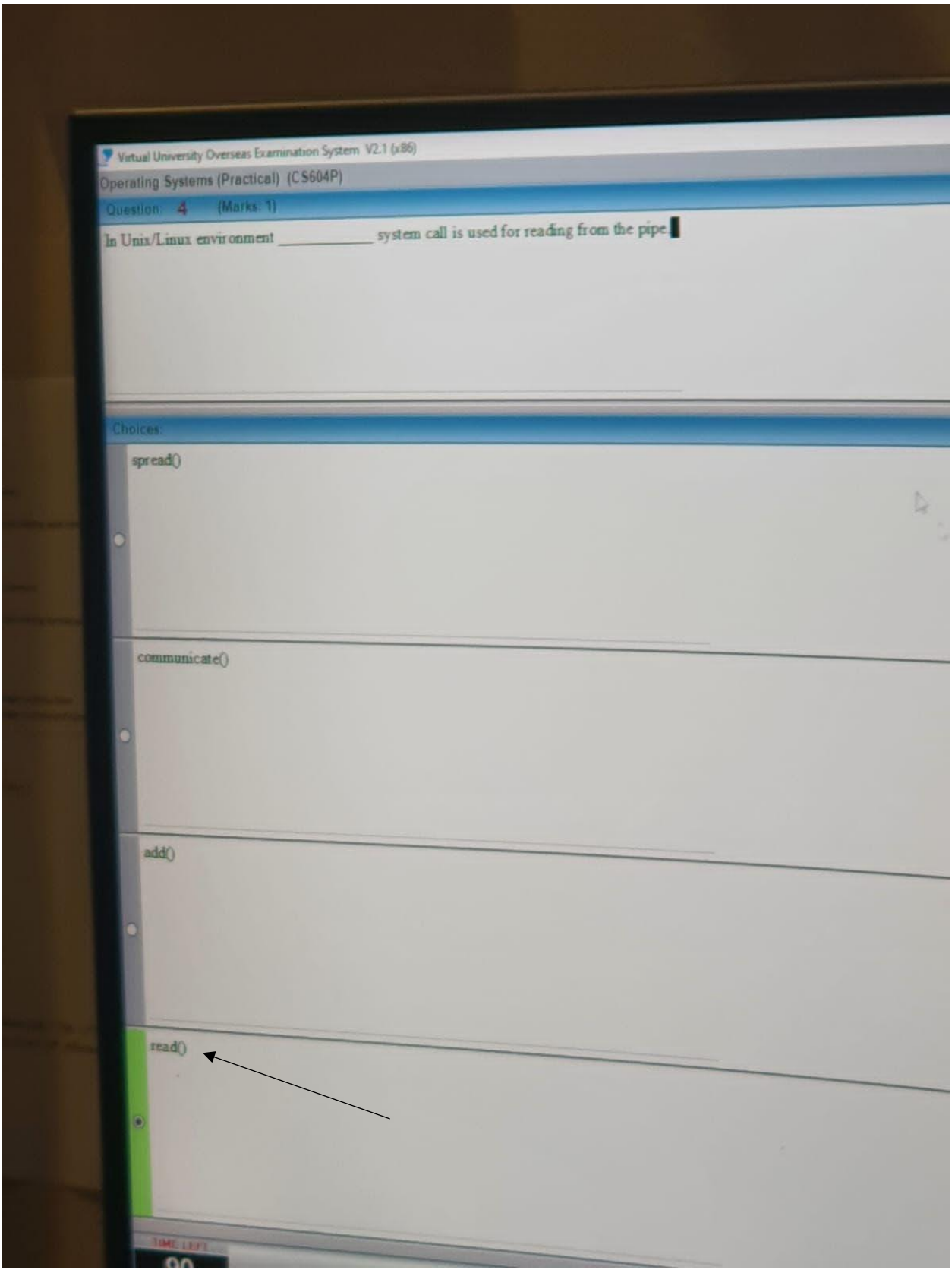
system()

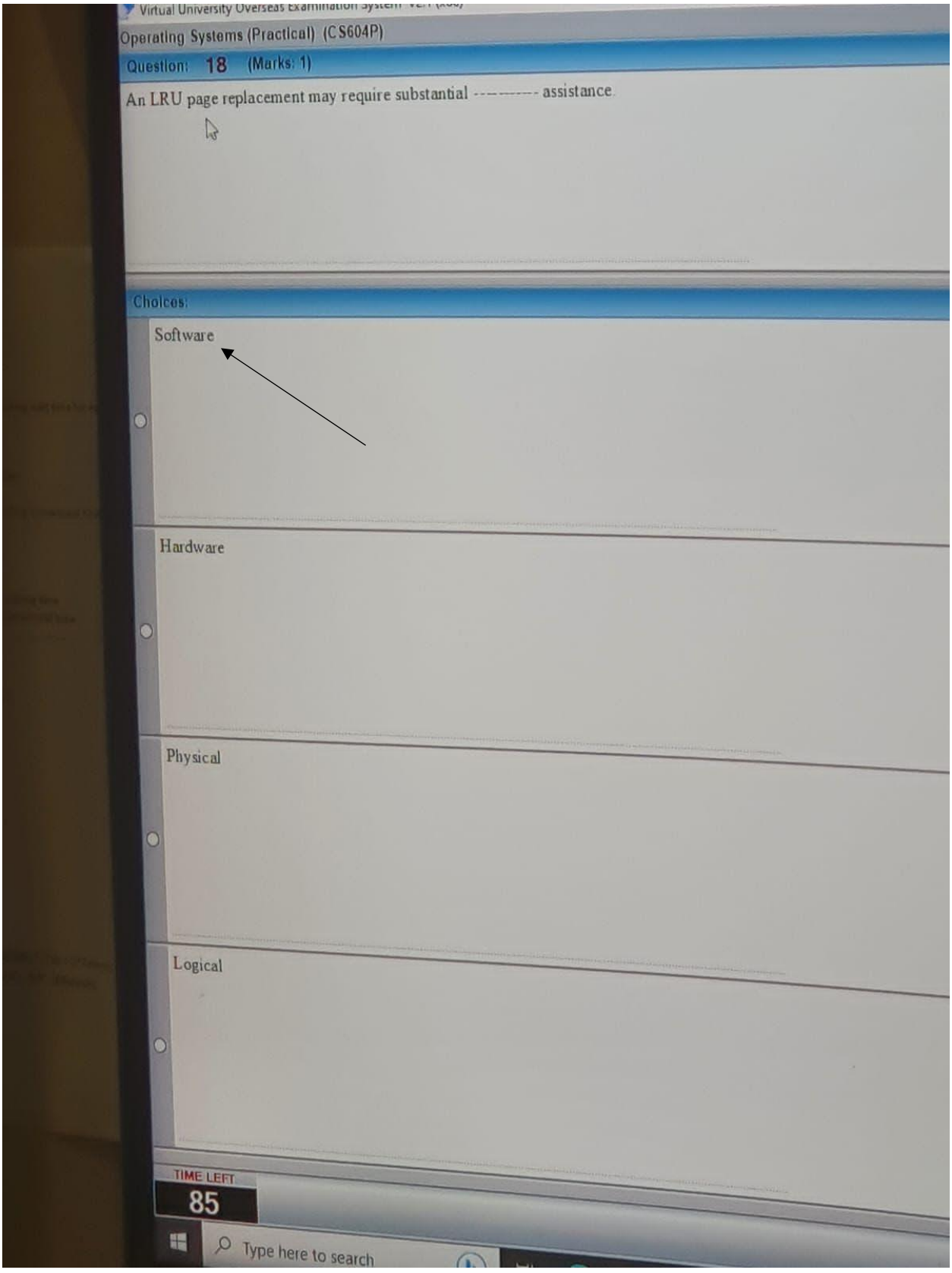
wait()

mkfifo()

fork()


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 5 (Marks: 1)
In Unix/Linux environment we used the _____ for communicating between the unrelated processes.
Choices:
fifo
system
lifo
stack


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 4 (Marks: 1)
In Unix/Linux environment _____ system call is used for reading from the pipe
Choices
spread()
communicate()
add()
read()
TIME LEFT


Virtual University Overseas Examination System
Operating Systems (Practical) (CS604P)
Question: 18 (Marks: 1)
An LRU page replacement may require substantial ---------- assistance.
Choices:
Software
Hardware
Physical
Logical
TIME LEFT
85
Type here to search

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (Practical) (CS604P)

Question: 14 (Marks: 1)

In the dining philosopher problem all the five philosophers sitting around the round table having _____chopsticks.

Choices:

6

5

4

7

TIME LEFT

86

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (Practical) (CS604P)

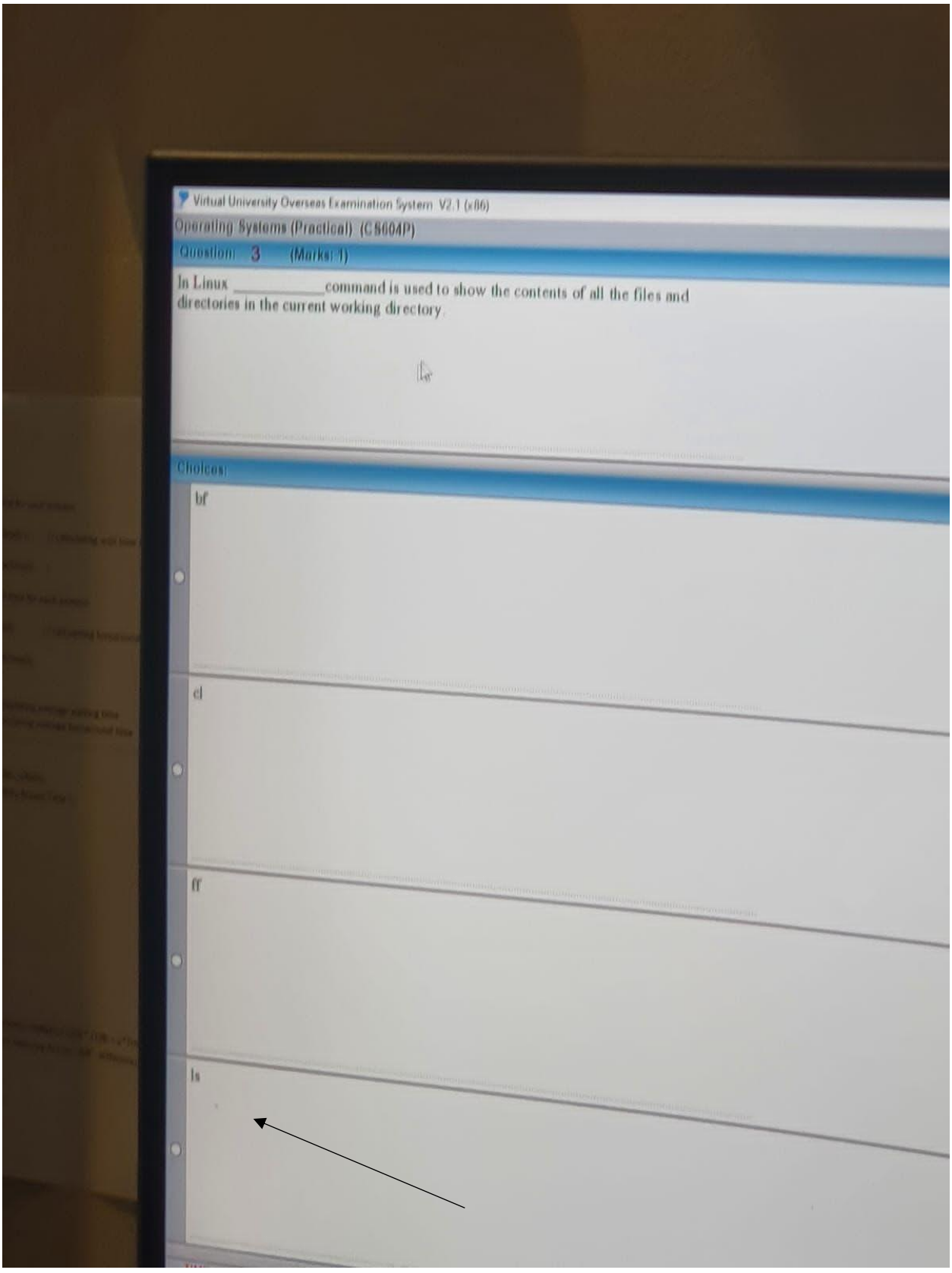
Question: 30 (Marks: 5)

Write down the formula along with the required parameters to calculate the overhead of bit map dealing with free space management also convert formula in C program.

Answer:

TIME LEFT
79

Type here to search


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 3 (Marks: 1)
In Linux _____command is used to show the contents of all the files and directories in the current working directory.
Choices:
bf
cl
ff
ls

Operating Systems (Practical) (CS604P)

Question 13 (Marks: 1)

In First Come First Served (FCFS) scheduling algorithm waiting times of P1 and P2 are 13 and 14 seconds respectively, then average waiting time will be.

Choices

13

13.5

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (Practical) (CS604P)

Question: 6 (Marks: 1)

In Unix/Linux environment we used the _____system call for creating the Fifo.

Choices:

create()

exit()

wait()

mknod()

Virtual University Overseas Examination System V2.1 (x86)

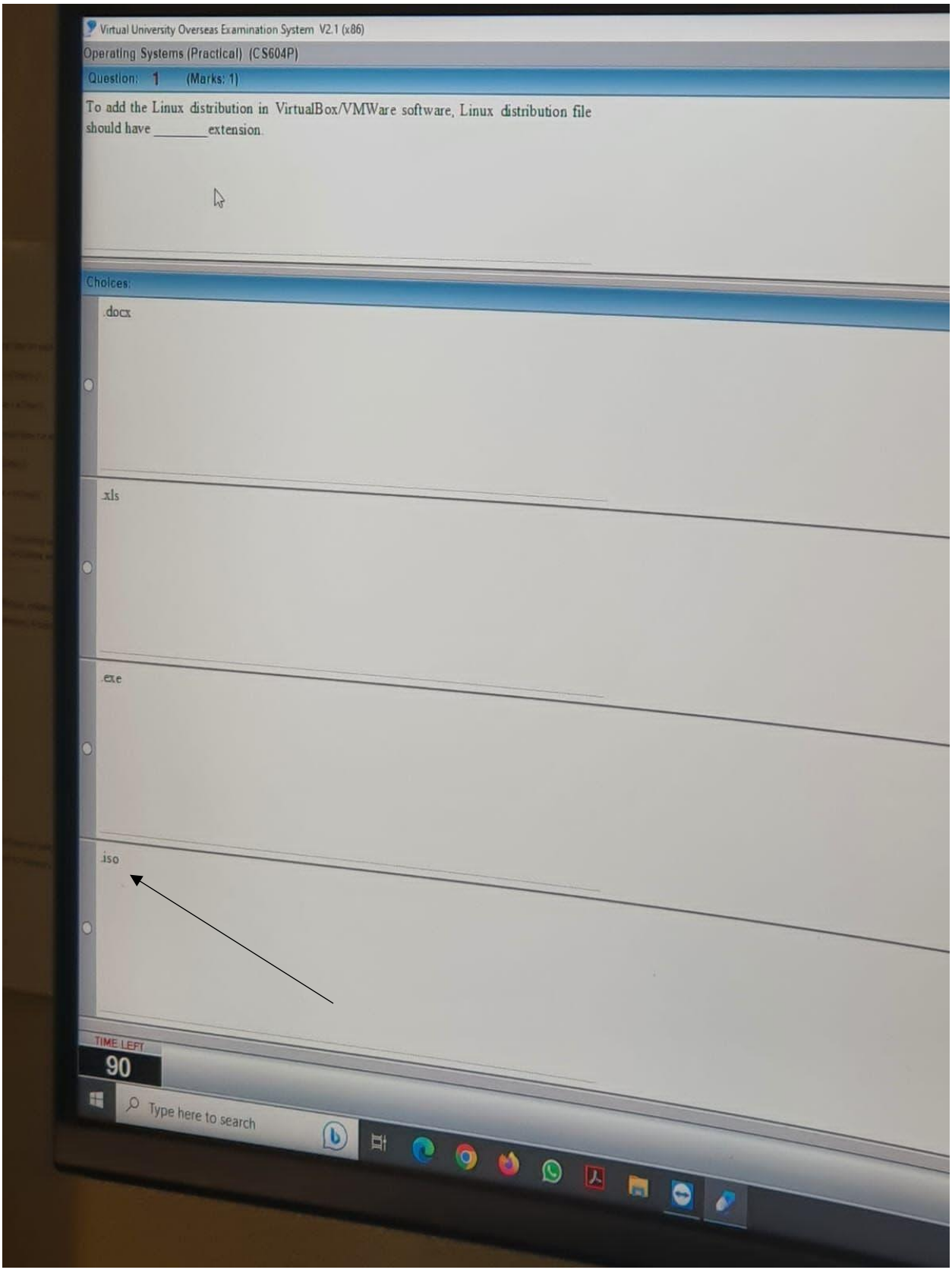
Operating Systems (Practical) (CS604P)

Question: 25 (Marks: 5)

Write down the complete calculation steps that convert 16 bit logical address into 20-bit physical address by considering the following parameters.

Logical address (16-bit)
IP = 0B10
CS = D000
Physical address (20-bit) =?

Answer:


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 1 (Marks: 1)
To add the Linux distribution in VirtualBox/VMWare software, Linux distribution file should have _____extension.
Choices:
.docx
.xls
.exe
.iso
TIME LEFT
90
Type here to search

Operating Systems (Practical) (CS604P)

Question: 12 (Marks: 1)

In Round Robin (RR) scheduling algorithm we have fixed time slot for every process.

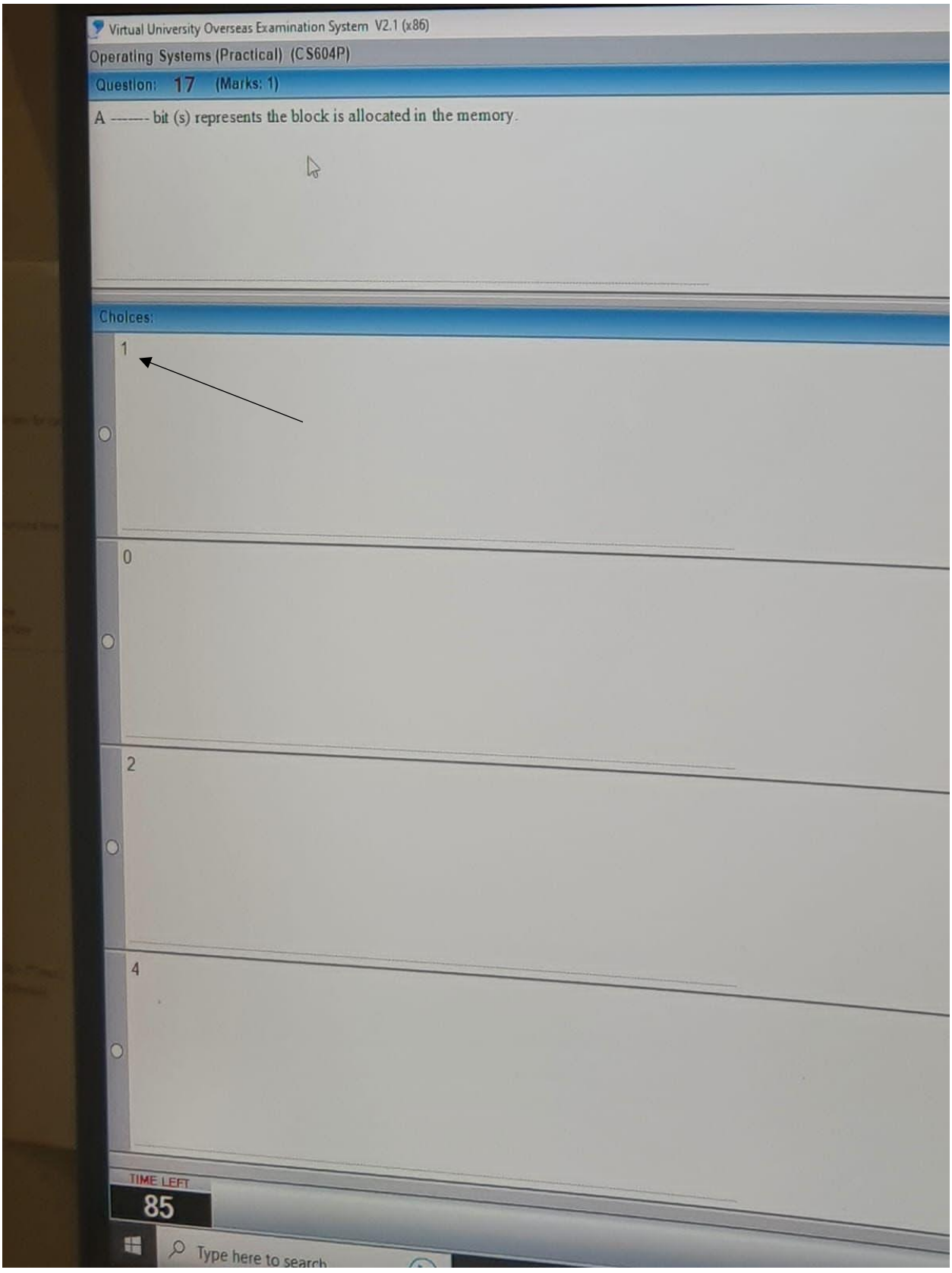
Choices:

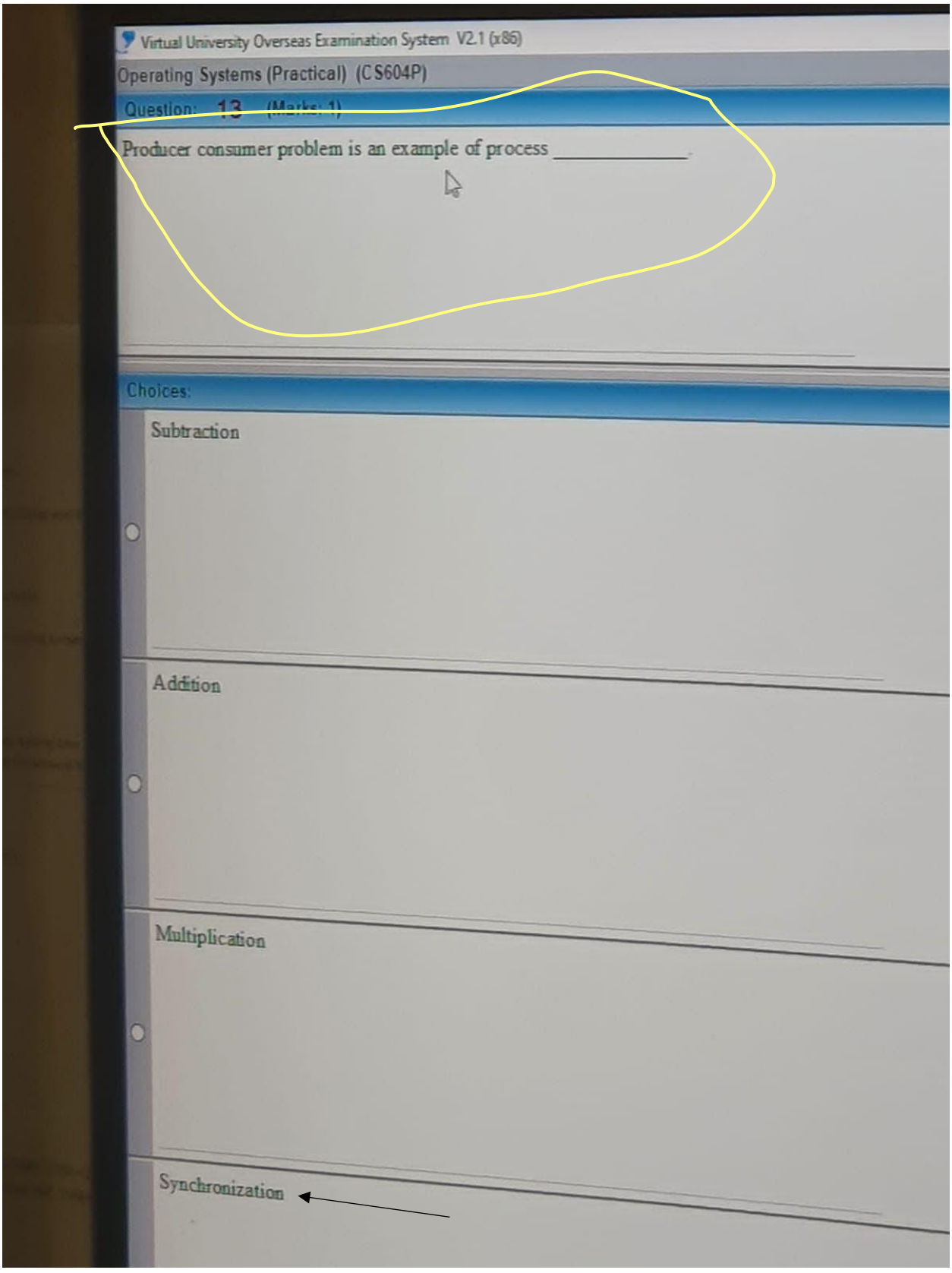
Non-Preemptive algorithm

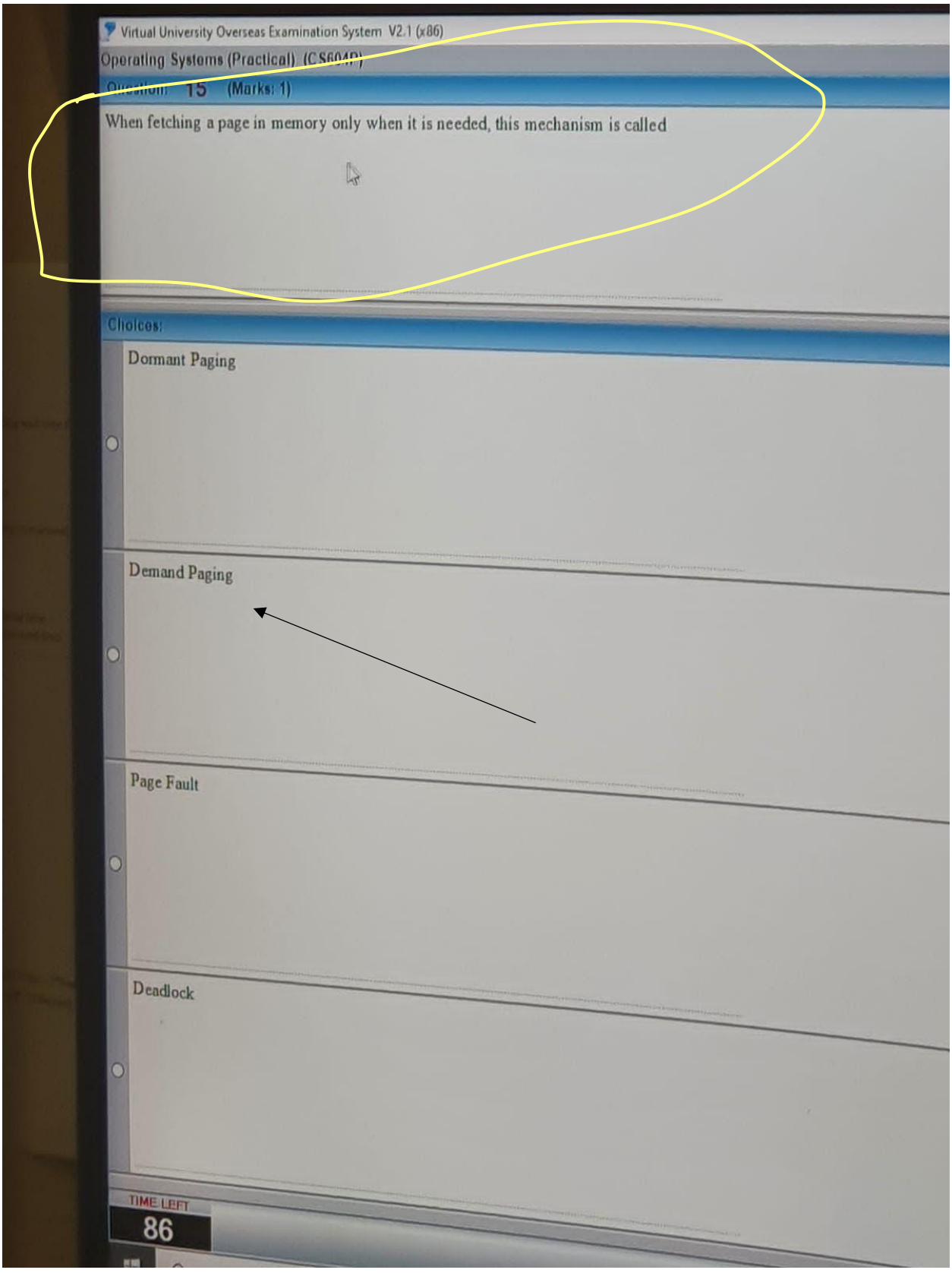
Preemptive algorithm

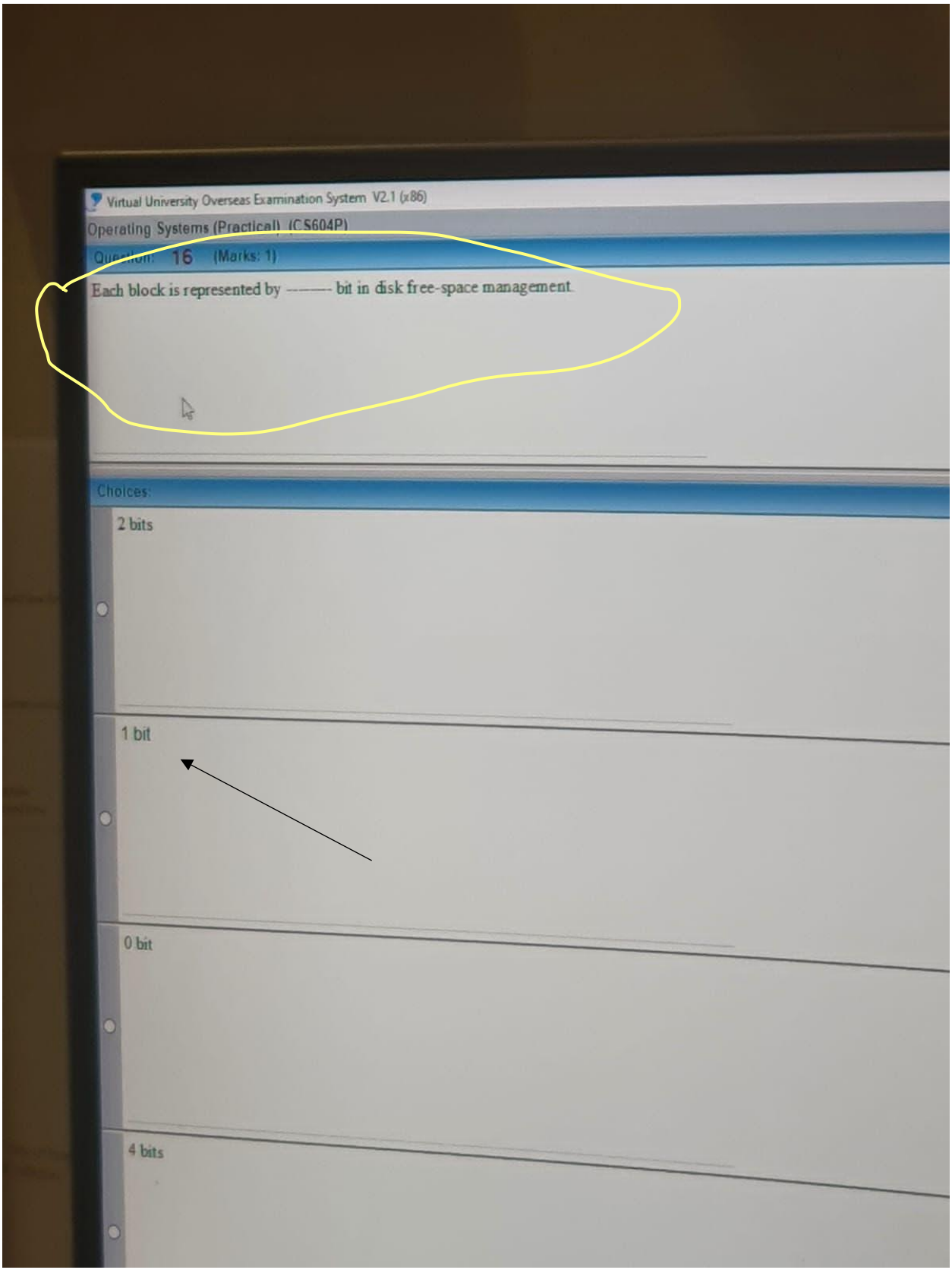
Subtractive algorithm

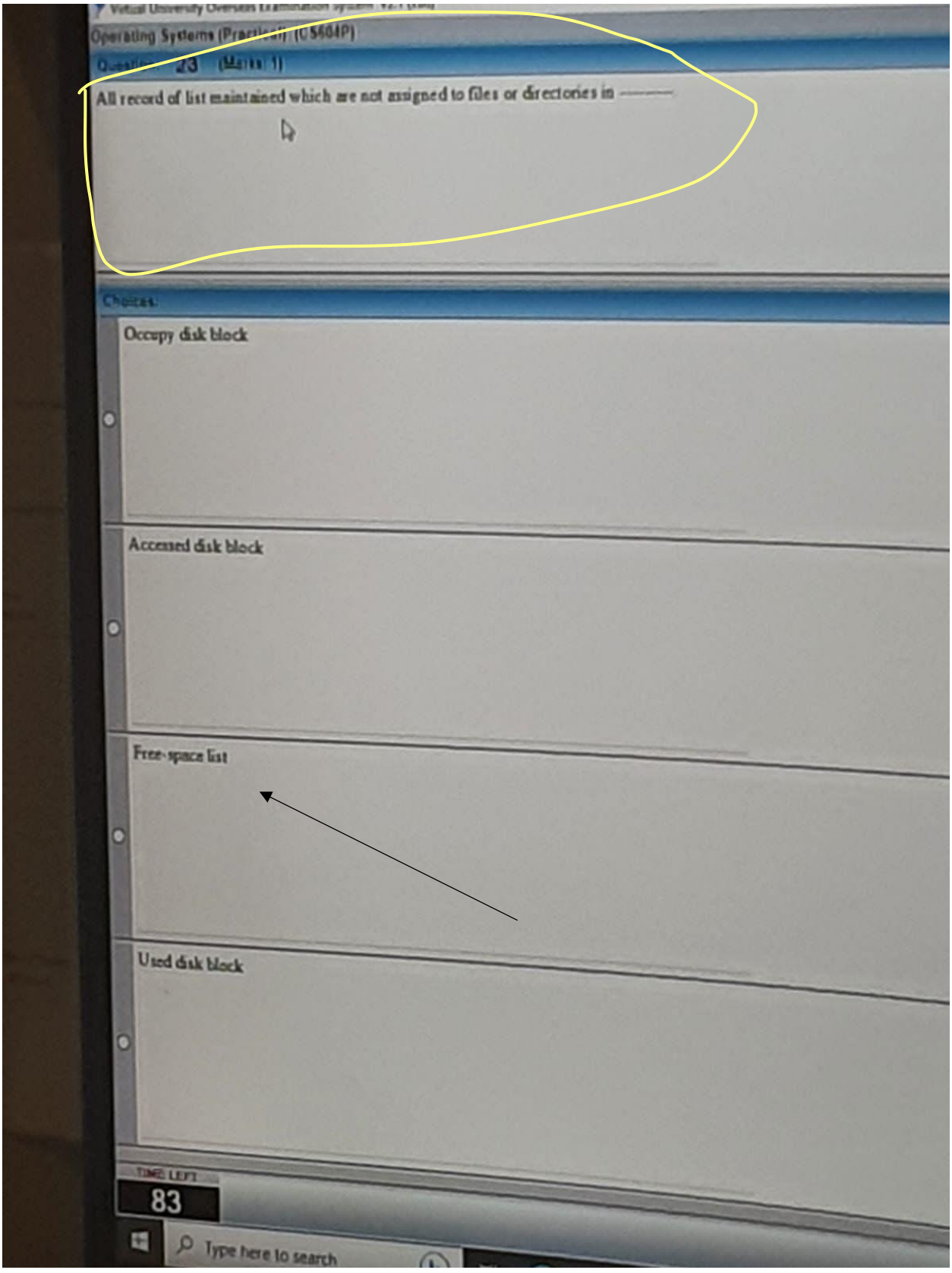
Additive algorithm

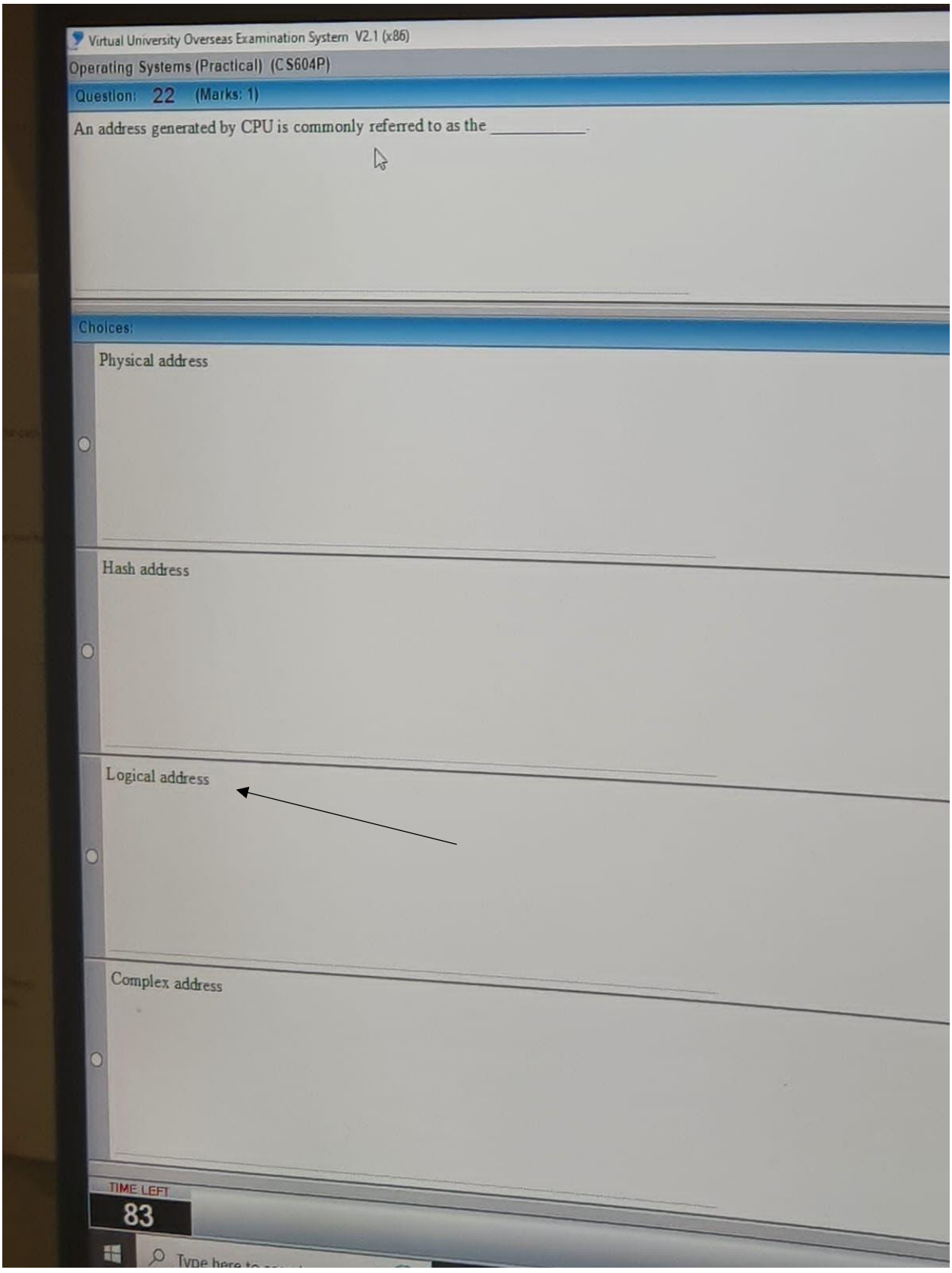

Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 17 (Marks: 1)
A ------ bit (s) represents the block is allocated in the memory.
Choices:
1
0
2
4
TIME LEFT
85
Type here to search

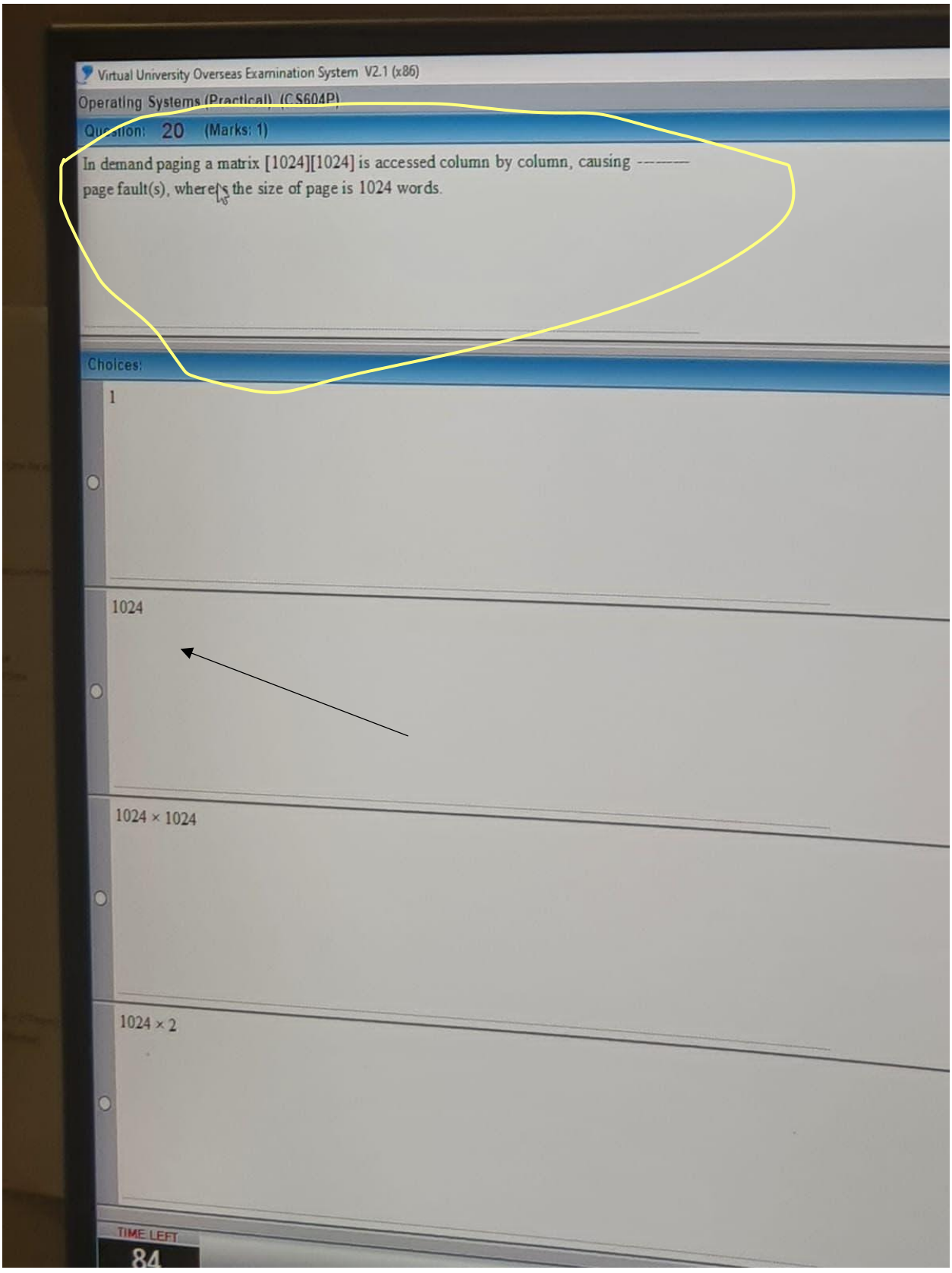

Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 13 (Marks: 1)
Producer consumer problem is an example of process _____.
Choices:
Subtraction
Addition
Multiplication
Synchronization


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 15 (Marks: 1)
When fetching a page in memory only when it is needed, this mechanism is called
Choices:
Dormant Paging
Demand Paging
Page Fault
Deadlock
TIME LEFT
86


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 16 (Marks: 1)
Each block is represented by ------- bit in disk free-space management
Choices:
2 bits
1 bit
0 bit
4 bits


Operating Systems (Practical) (CS604P)
23 (Marks: 1)
All record of list maintained which are not assigned to files or directories in ———
Occupy disk block
Accessed disk block
Free-space list
Used disk block
TIME LEFT
83
Type here to search


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 22 (Marks: 1)
An address generated by CPU is commonly referred to as the _____.
Choices:
Physical address
Hash address
Logical address
Complex address
TIME LEFT
83


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 20 (Marks: 1)
In demand paging a matrix [1024][1024] is accessed column by column, causing ------- page fault(s), where's the size of page is 1024 words.
Choices:
1
1024
1024 × 1024
1024 × 2
TIME LEFT
84

Question 24 (Marks 1)

In disk space management a bit vector is implemented as a _____

Choices

Ready list

Free space list

Waiting list

User space list

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (Practical) (CS604P)

Question: 26 (Marks: 5)

For Inter-process communication using pipes, between related process read() and write() system calls are used. You are required to write the syntax of read() and write() system calls.

Answer:

B I U

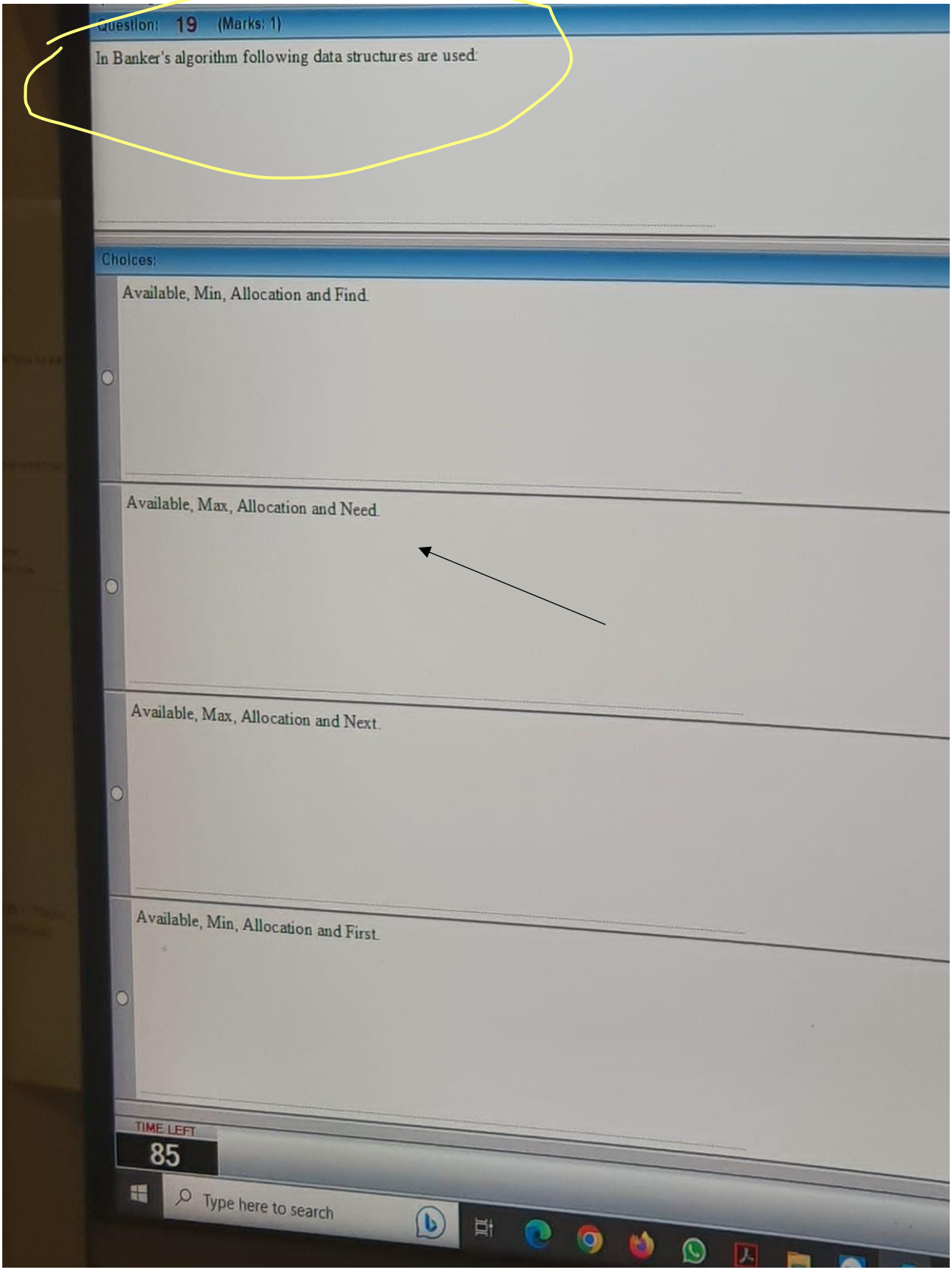
TIME LEFT

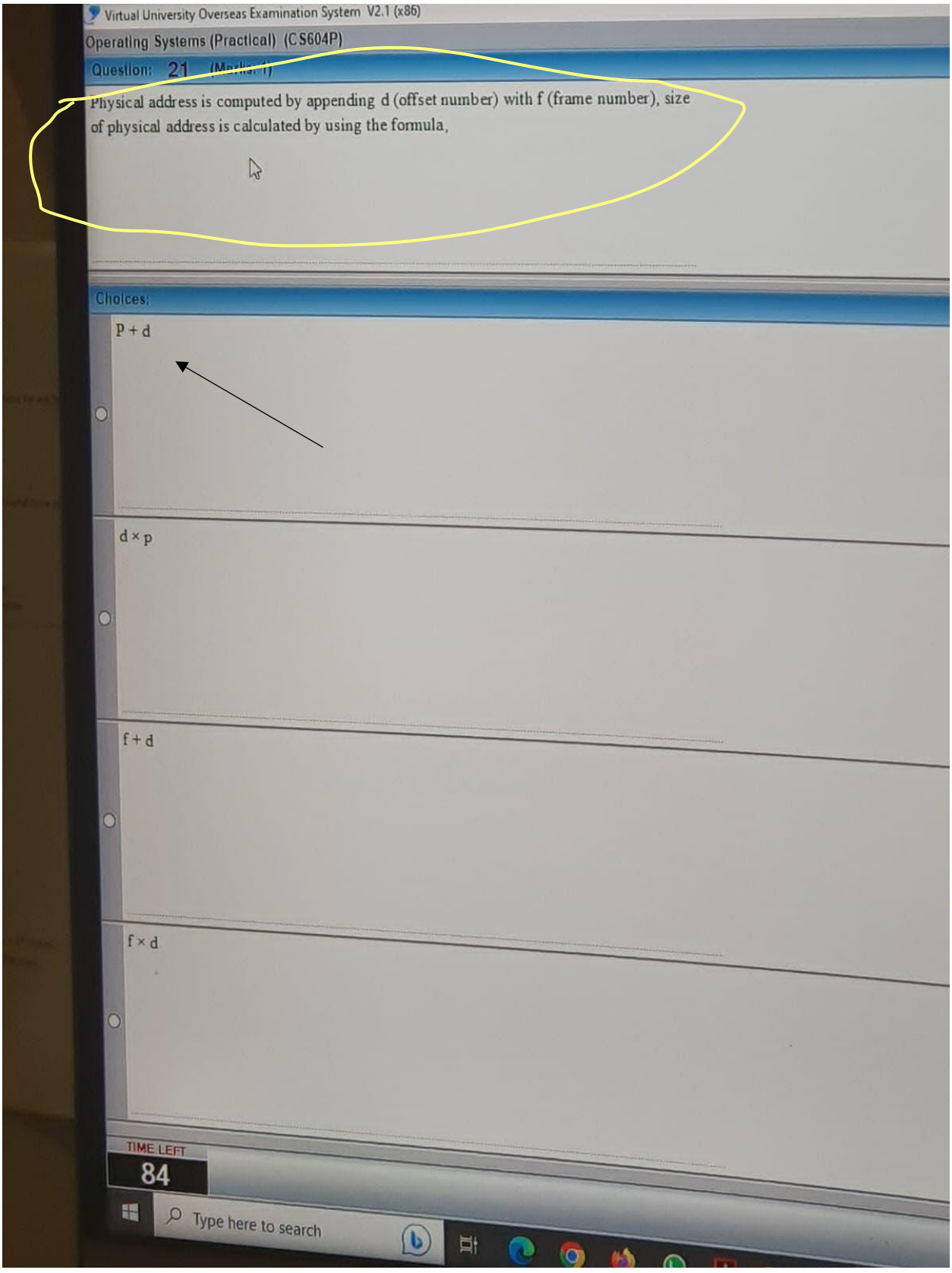
Question: 11 (Marks: 1)

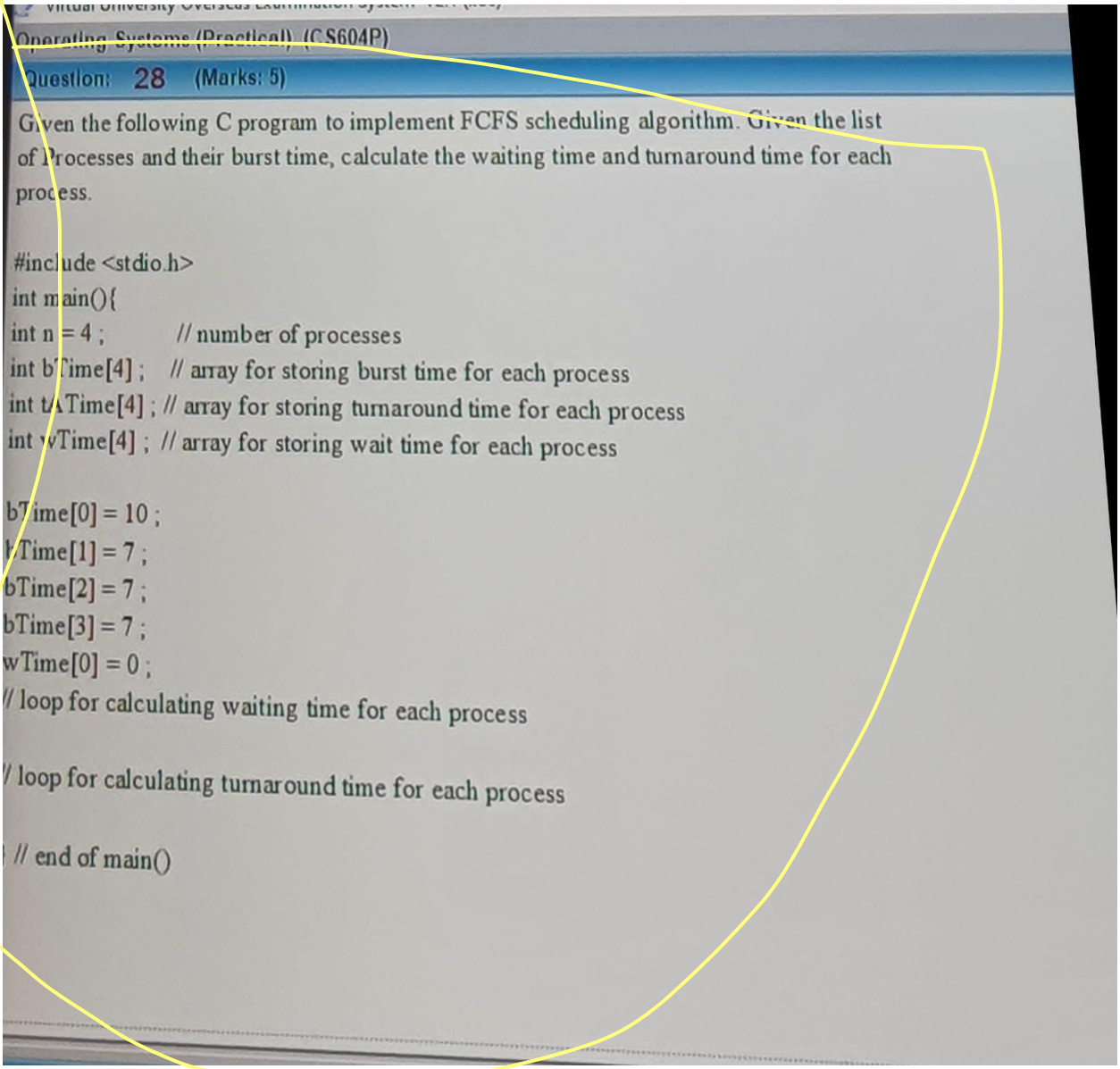
In _____scheduling algorithm we have fixed time slot for every process.

Choices:

- Decision Tree
- Round Robin
- Random Forest
- Naive Bayes


Question: 19 (Marks: 1)
In Banker's algorithm following data structures are used
Choices:
Available, Min, Allocation and Find.
Available, Max, Allocation and Need.
Available, Max, Allocation and Next.
Available, Min, Allocation and First.
TIME LEFT
85
Type here to search


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 21
Physical address is computed by appending d (offset number) with f (frame number), size of physical address is calculated by using the formula,
Choices:
P + d
d × p
f + d
f × d
TIME LEFT
84
Type here to search



Operating Systems (Practical) (CS604P)

Question: 28 (Marks: 5)

Given the following C program to implement FCFS scheduling algorithm. Given the list of Processes and their burst time, calculate the waiting time and turnaround time for each process.

```
#include <stdio.h>
int main(){
int n = 4 ;          // number of processes
int bTime[4] ;   // array for storing burst time for each process
int tATime[4] ; // array for storing turnaround time for each process
int wTime[4] ;  // array for storing wait time for each process

bTime[0] = 10 ;
bTime[1] = 7 ;
bTime[2] = 7 ;
bTime[3] = 7 ;
wTime[0] = 0 ;
// loop for calculating waiting time for each process

// loop for calculating turnaround time for each process

} // end of main()
```

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (Practical) (CS604P)

Question: 29 (Marks: 5)

synchronization. A producer can produce one item while the consumer is consuming another item. The producer and consumer must be synchronized so that the producer does not try to produce an item when the buffer is full. Otherwise, it shall store the item in the buffer. You are required to complete the producer code.

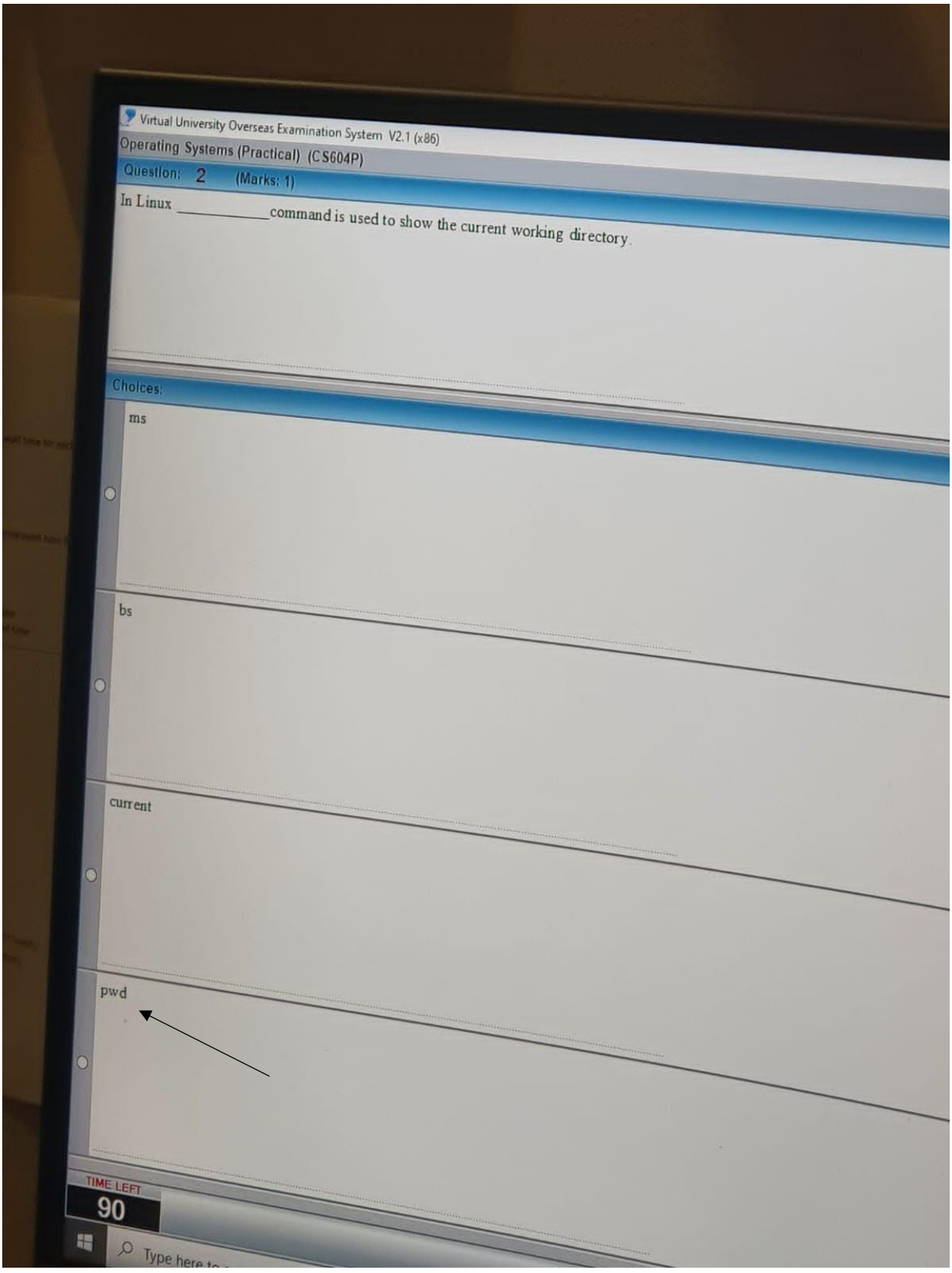
```
#include<stdio.h>
int main()
{
int bufsize, in, out, produce = 0, consume = 0 , counter = 0;
in = 0; out = 0;

printf("Please enter the size of buffer: ");
scanf("%d", &bufsize);
int buffer[bufsize] ;
char choice ;
printf ("Enter p to produce an item in buffer:\n") ;
printf ("Enter q to Quit:\n") ;
do{
scanf("%c", &choice);
switch(choice){
case 'p':
// code for producer
break ;
}
} // end of switch
}while( choice != 'q') ;
}
```

Answer:

TIME LEFT

79

Type here to search


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: 2 (Marks: 1)
In Linux _____command is used to show the current working directory.
Choices:
ms
bs
current
pwd
TIME LEFT
90
Type here to

Virtual University Overseas Examination System

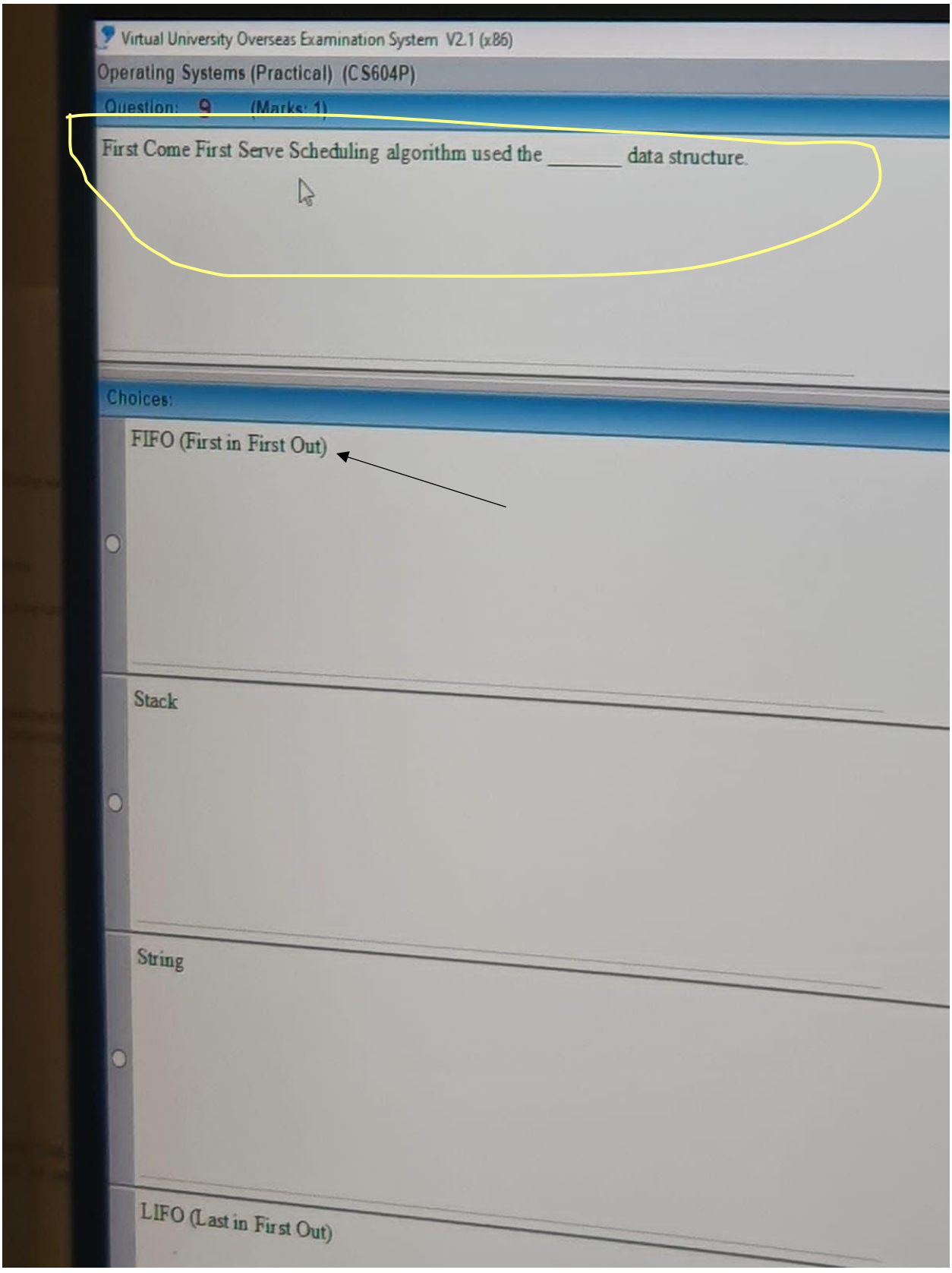
Operating Systems (Practical) (CS604P)

Question: 10 (Marks: 1)

In First-Come-First-Served (FCFS) scheduling algorithm waiting times of P1 and P2 are 13 and 14 seconds respectively, their average waiting time will be.

Choices:

- 10
- 15
- 16
- 13.5


Virtual University Overseas Examination System V2.1 (x86)
Operating Systems (Practical) (CS604P)
Question: (Marks: 1)
First Come First Serve Scheduling algorithm used the _____ data structure.
Choices:
FIFO (First in First Out)
Stack
String
LIFO (Last in First Out)

Question: 8 (Marks: 1)

In the First Come First Serve Scheduling algorithm turnaround time for each process can be calculated by:

Choices:

- Square root of waiting time
- Squaring the average waiting time
- Adding waiting time and burst time
- Multiplying the waiting and average waiting time

Virtual University Overseas Examination System V2.1 (x86)

Operating Systems (Practical) (CS604P)

Question: 27 (Marks: 5)

Given the following C program to implement FCFS scheduling algorithm. Given the list of Processes and their burst time, calculate the waiting time for each process and then displays it for each process. Also, calculate and display average waiting time.

```
#include <stdio.h>
int main(){
int n = 4 ;          // number of processes
int bTime[4] ;   // array for storing burst time for each process
int wTime[4] ; // array for storing wait time for each process
int tATime[4] ; // array for storing turnaround time for each process
bTime[0] = 10 ;
bTime[1] = 5 ;
bTime[2] = 6 ;
bTime[3] = 9 ;
int avgWTime = 0 ;
wTime[0] = 0 ;
// loop for calculating waiting time for each process

// calculating average waiting time

// loop for displaying waiting time
// displaying average waiting time

} // end of main()
```

Answer: